Bachelor of Computer Applications

Semester – V

Business Analytics

Assignment No.: 4

Title: Data Cleaning using Power Query

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Date: 09/09/26

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Data Cleaning using Power Query

Problem Statement

A company has received customer data containing missing values, duplicate records, and inconsistent entries. Clean and prepare the dataset using Power Query before analysis.

Dataset Link:

https://www.kaggle.com/datasets/ahmedmohamed2003/cafe-sales-dirty-data-for-cleaning-training

Methodology / Steps Followed

Step 1: Data Collection

· Obtained the Cafe Sales Data from the CSV file downloaded from **Kaggle**.

· Used the **dirty_cafe_sales.csv** dataset for data cleaning and analysis.

Step 2: Data Import

· Opened Power BI Desktop.

· Imported the CSV dataset using the **Get Data** option.

· Selected **Transform Data** to open the dataset in Power Query Editor.

Step 3: Data Cleaning and Transformation

· Checked the available columns and removed unnecessary top rows.

· Renamed the required columns for better understanding.

· Removed duplicate records from the dataset.

· Removed rows containing errors and missing values.

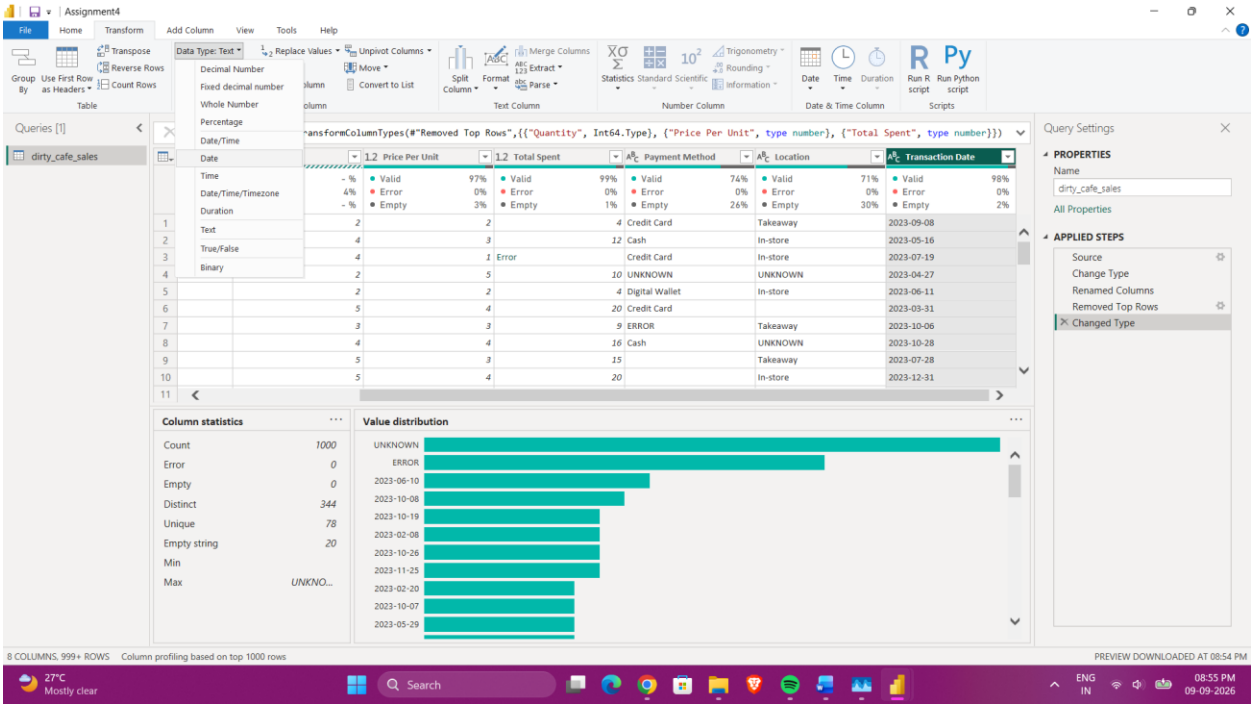
Step 4: Data Type Correction and Filtering

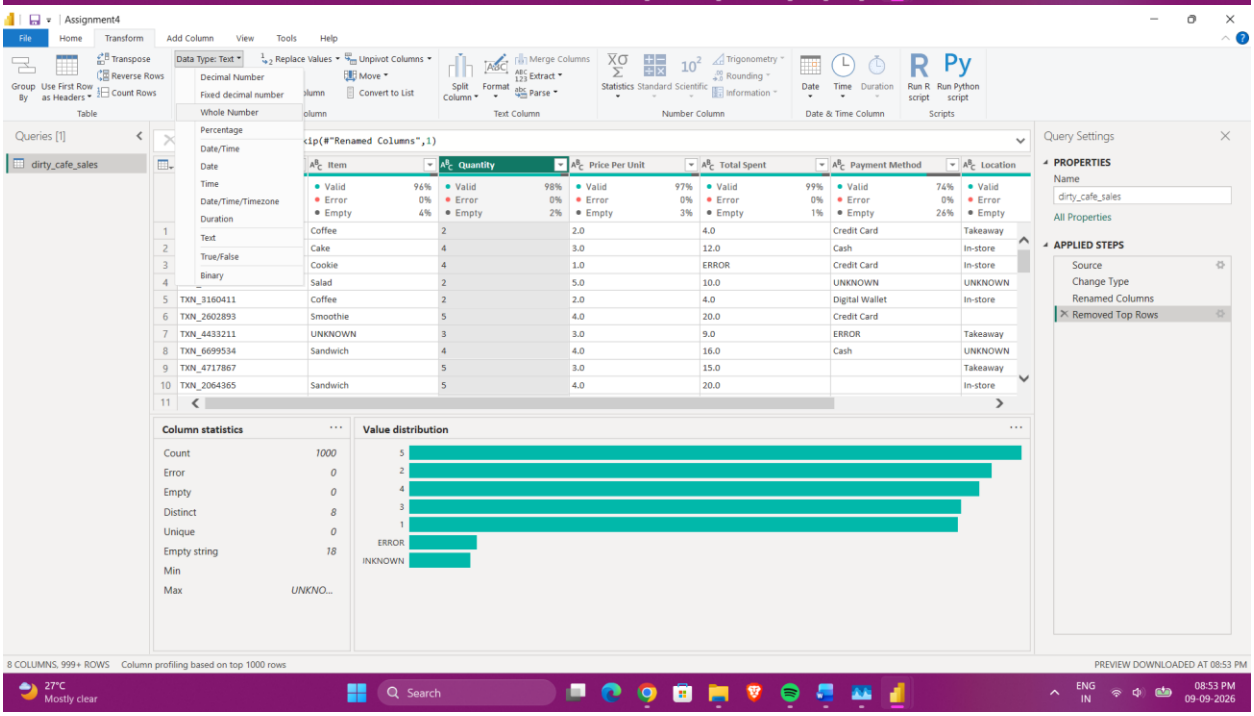
· Changed the **Quantity** column to Whole Number.

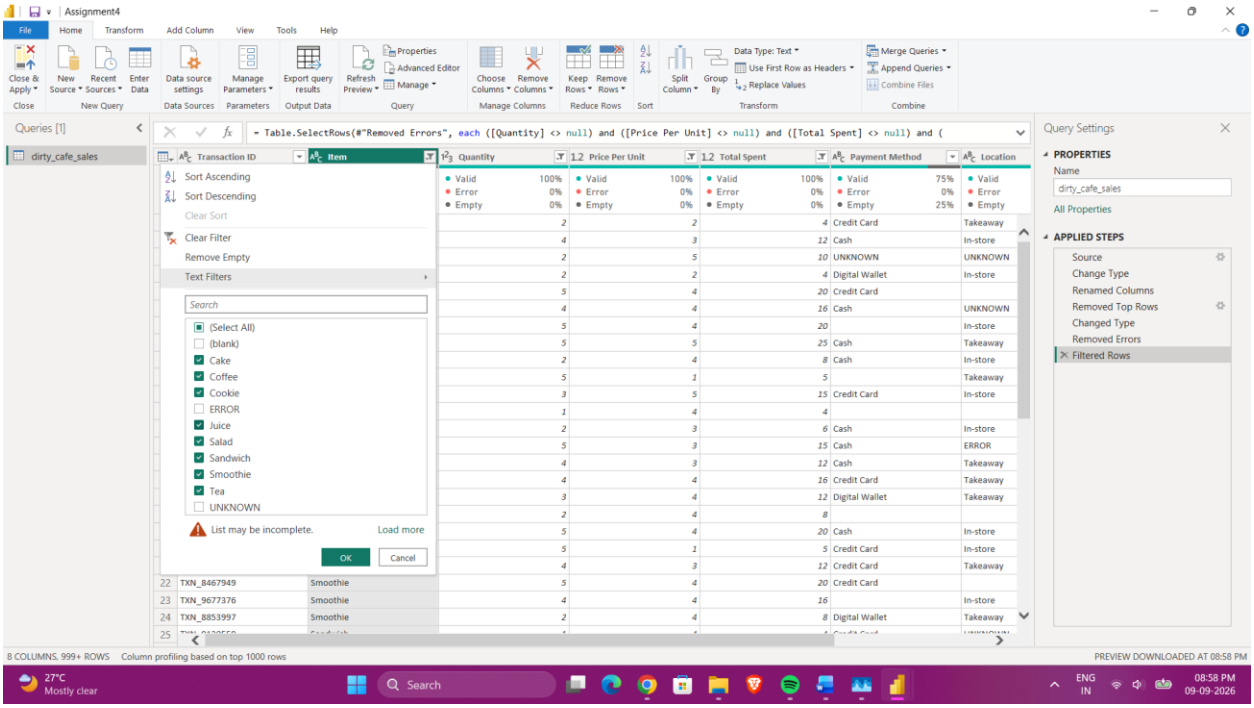
· Changed **Price Per Unit** and **Total Spent** to Decimal Number.

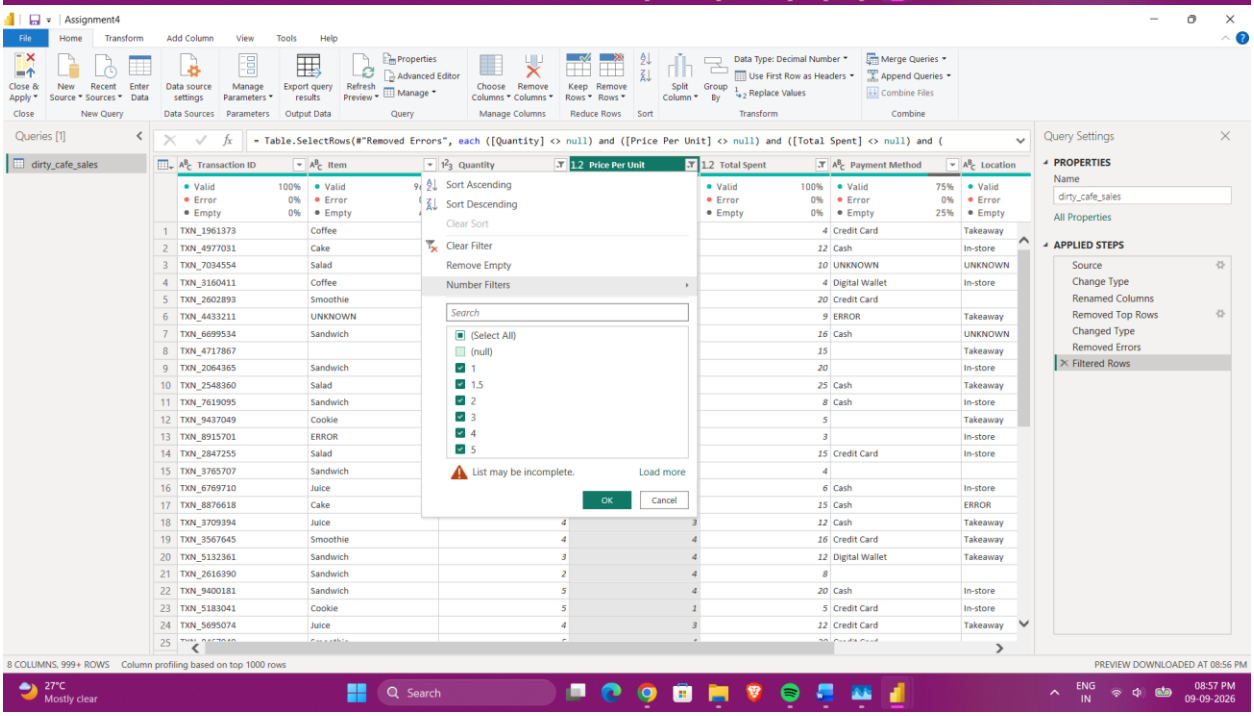
· Changed the **Transaction Date** column to Date format.

· Filtered out invalid entries such as **ERROR** and **UNKNOWN** from the required fields.


Assignment4
File Home Transform Add Column View Tools Help
Data Type: Text
Decimal Number
Fixed decimal number
Whole Number
Percentage
Date/Time
Date
Time
Date/Time/Timezone
Duration
Text
True/False
Binary
Queries [1]
dirty_cafe_sales
Transaction ID | Item | Quantity | Price Per Unit | Total Spent | Payment Method | Location | Transaction Date
Column statistics
Count 1000
Error 0
Empty 0
Distinct 344
Unique 78
Empty string 20
Min
Max UNKNO...
Value distribution
UNKNOWN
ERROR
2023-06-10
2023-10-08
2023-10-19
2023-02-08
2023-10-26
2023-11-25
2023-02-20
2023-10-07
2023-05-29
Query Settings
PROPERTIES
Name
dirty_cafe_sales
All Properties
APPLIED STEPS
Source
Change Type
Renamed Columns
Removed Top Rows
Changed Type
8 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 08:54 PM
27°C Mostly clear
Search
ENG IN
08:55 PM 09-09-2026


Assignment4
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Date/Time/Timezone
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Queries [1]
dirty_cafe_sales
Transaction ID | Item | Quantity | Price Per Unit | Total Spent | Payment Method | Location
Column statistics
Count 1000
Error 0
Empty 0
Distinct 8
Unique 0
Empty string 18
Min
Max UNKNO...
Value distribution
5
2
4
3
1
ERROR
INKNOWN
Query Settings
PROPERTIES
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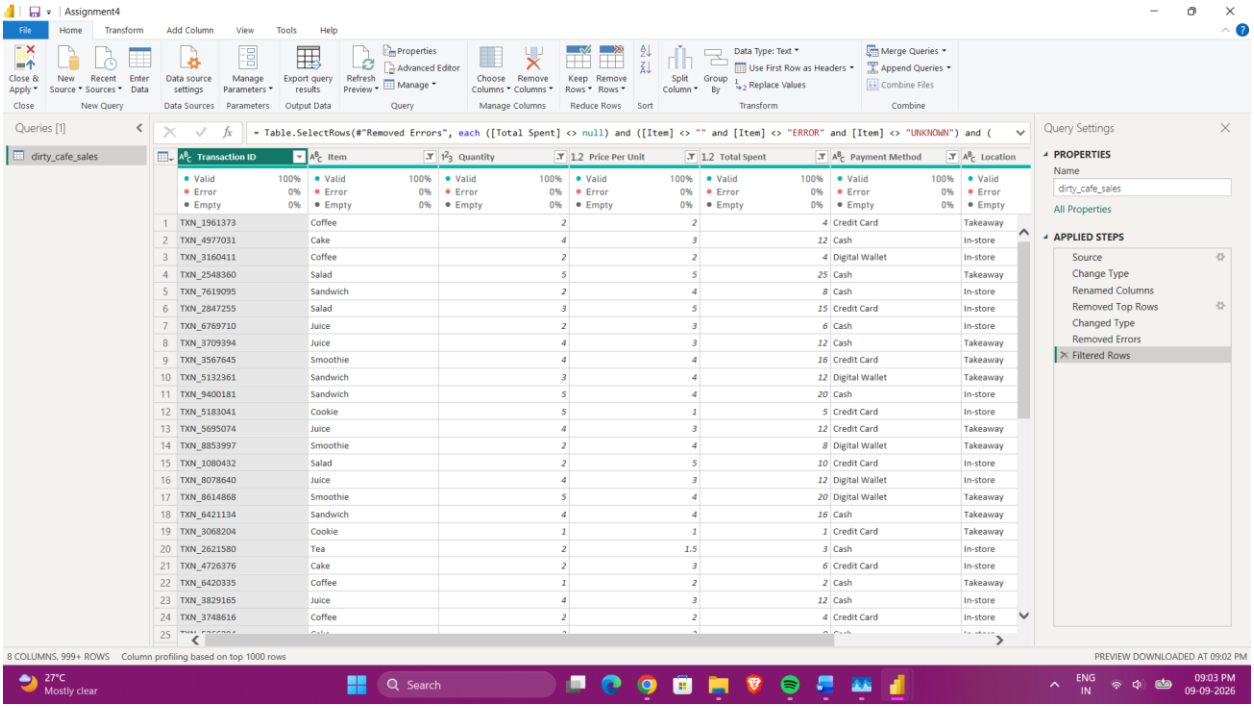
Step 5: Data Quality Check

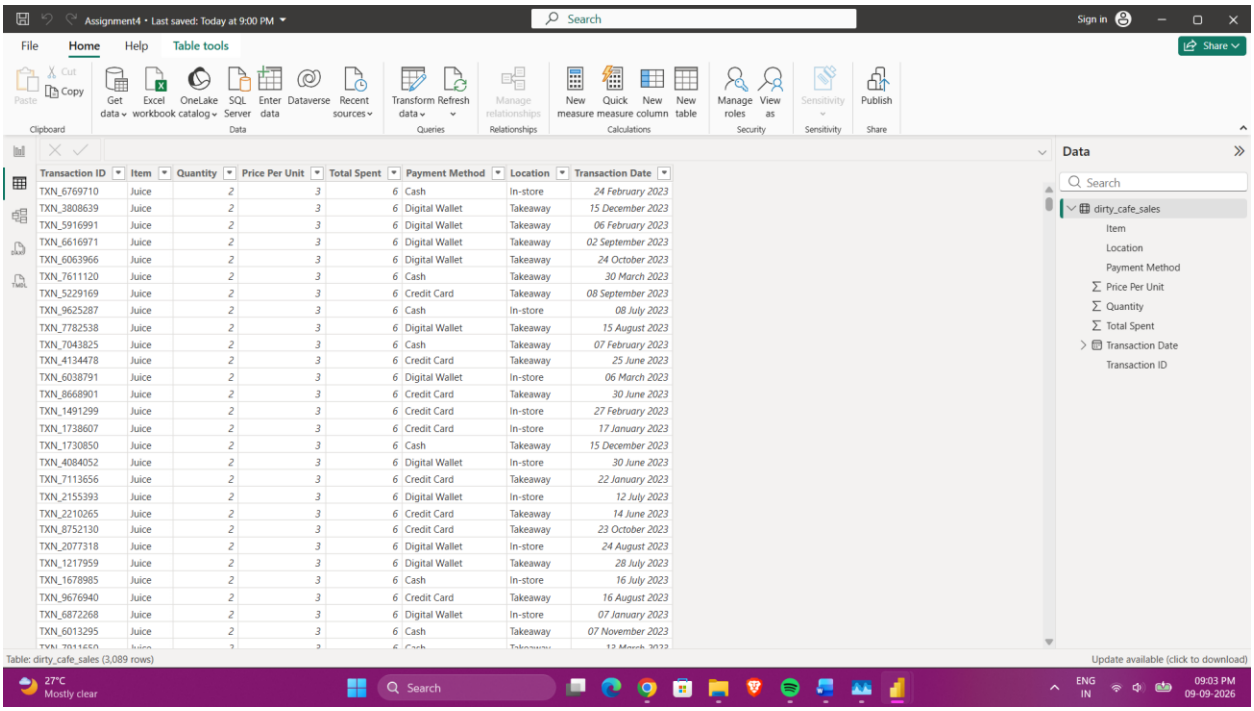
· Used **Column Quality** to check the cleaned dataset.

· Verified that the data contains valid values with no errors or empty values.

· Prepared the cleaned dataset for further analysis in Power BI.

Output





The final output provides:

· Cleaned Cafe Sales Data

· Removed Duplicate Records

· Corrected Data Types

· Removed Error Values

· Removed Missing Values

· Removed Inconsistent Entries

· Valid and Consistent Dataset

The cleaned data is now ready for further analysis in Power BI.

Conclusion

This project successfully cleaned and prepared the Cafe Sales data using Power Query in Power BI. The data was organized and transformed by removing duplicate records, errors, missing values, and inconsistent entries. The data types were also corrected to make the dataset suitable for analysis. The final cleaned dataset provides valid and consistent data, helping users perform better analysis and make informed decisions.

Learning Outcomes

Through this project, I learned:
· How to import CSV files into Power BI.
· How to use Power Query for data cleaning and transformation.
· How to identify and remove duplicate records from a dataset.
· How to remove errors and missing values from the data.
· How to correct data types for different columns.
· How to identify and remove inconsistent entries such as ERROR and UNKNOWN.
· How to use Column Quality to check the quality of cleaned data.
· How to prepare a clean and consistent dataset for further analysis in Power BI.

Github - https://github.com/Syahrulnalli/Business-Analytics